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Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
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Task Description:	Astronauts in deep-space missions are exposed to galactic cosmic radiation, which consists of HZE particles. Both the1996 National Academy of Science report of interplanetary mission crew hazards and the NASA Radiation Health Bioastronautics Roadmap emphasize that estimating radiation risks to the central nervous system is a high research priority. One CNS cell population that warrants careful analysis for their response to radiation is adult neural stem cells, since these pluripotent cells may play an important role in recovery after radiation-induced brain injury. Studies from radiation biology and tumor biology have generated mixed information whether adult neural stem cells are radiosensitive or not. Understanding how adult neural stem cells respond to radiation and identification of the characteristics of the radioresistant neural stem cells will be pivotal in crafting future strategies to harness stem cells for repair of the injured brain. In this proposed study, We will perform a comprehensive analysis of the response and potential recovery of adult neural stem cells after HZE radiation using two mouse models: 1) the nestinGFP transgenic mouse, which allows visualization of the neural stem cell population within the adult brain at any point in time; and 2) our unique nestin-CreERT2/R26R-YFP transgenic mouse, which allows inducible and permanent YFP labeling of adult neural stem cells and their progeny for the remainder of the cell's life. These complementary models allow us to track the proliferation, differentiation, and survival of neural stem cells and their progeny in vivo after HZE radiation. We hypothesize that adult neural stem cells maintain some ability to repopulate the injured brain after radiation, and that this ability improves with time. We further hypothesize that these invulnerable stem cells will provide clues to radioresistance, which may involve the multiple strategies that neural stem cells employ to repair damaged DNA.		
Rationale for HRP Directed Research:			
Research Impact/Earth Benefits:	The proposed research targets several questions on NASA's Radiation Health Bioastronautics Roadmap, all in the category of Acute and Late CNS Risks. The proposed research utilizes two novel transgenic mouse models for a comprehensive analysis of adult neural stem cells and their progeny in vivo for the temporal events of cells survival, proliferation, and differentiation in response various low- or high-LET radiation. The resulting information on the long-term neurogenic and/or degenerative activities will provide accurate assessments of CNS risks after space radiation exposure. The results from Aim 3 in particular will provide strong indication whether adult neural stem cells can even partially recover from radiation injury and produce mature neurons. This approach can also identify the threshold of radiation injury to adult neurogenic potential. Understanding how adult neural stem cells respond to radiation will allow assessment of the tantalizing possibility that the remaining neural stem cells to repopulate the more rapidly dividing progenitor cells and immature neurons that are sensitive to radiation. Even if the remaining neural stem cells are not able to repopulate the cell types lost after irradiation, identification of the characteristics of the radioresistant neural stem cells will itself be pivotal in crafting future strategies to harness stem cells for repair of the injured brain.		
	Galactic cosmic radiation (GCR) consists of high-LET HZE particles and poses tremendous health risks to astronauts, including cognitive impairment. The neural basis for cognitive deficits is unclear, leading both the1996 National Academy of Science report of interplanetary mission crew hazards and the NASA Radiation Health Bioastronautics Roadmap to make estimation of radiation risks to the central nervous system (CNS) a high research priority. One CNS cell population that warrants close analysis for their response to radiation is adult neural stem cells, as these pluripotent cells may play a key role in cellular and cognitive recovery after brain injury. Studies from radiation and tumor biology have generated mixed information whether adult neural stem cells are radiosensitive or not. Understanding how adult neural stem cells respond to radiation, and identifying characteristics of potentially radioresistant neural stem cells will be pivotal in crafting future strategies to harness stem cells for repair of the injured brain and prevention of GCR-induced CNS injury. We propose a comprehensive analysis of the response – and potential recovery – of adult neural stem cells in vivo after low- and high-LET irradiation. We hypothesize that neural stem cells and their progeny will be differentially vulnerable to radiation injury, including DNA damage. We propose three aims to determine: Aim 1. The impact of low- and high-LET irradiation on adult neural stem cells. Stem cells in our nestinGFP transgenic mouse are identifiable based on green fluorescent protein, morphology, and phenotype. We will exploit this ability to identify stem cells in vivo by subjecting nestinGFP mice to low-LET X-ray and various high-LET HZE particle radiation. We will study the long-term effect on adult neural stem cell proliferation and differentiation (1) in vivo and (2) in vitro. Aim 2. The impact of low- and high-LET irradiation on the progeny of adult neural stem cells. Our nestin-CreERT2/R26R-YFP mouse allows permanent YFP-labeling of adult neural stem cells and their progeny. We will exploit this ability to track stem cells and their progeny throughout life by first inducing YFP-labeling of adult neural stem cells and their progeny in nestin-CreERT2/R26R-YFP mice and then subjecting the mice to low-LET X-ray and various high-LET HZE particle radiation. We will study the long-term effect of radiation on (1) YFP+ neural stem cells in vivo and (2) YFP+ progeny in vivo.		

	Aim 3. The neurogenic potential of adult neural stem cells after radiation injury. We will subject nestin-CreERT2/R26R-YFP mice to low-LET X-ray radiation and various high-LET HZE particle radiation. After radiation exposure and brief recovery, we will then induce permanent YFP-labeling in the surviving adult neural stem cells to investigate the dosage effect of low- and high-LET irradiation on the neurogenic potential of YFP+ adult neural stem cells (1) in vivo and (2) in vitro. Taken together, these aims will allow comprehension of radiation-induced injury to stem cells and their progeny, and thus will lay the critical groundwork for assessing and ultimately preventing GCR-induced injury.
Task Progress:	Summary of the results obtained: During the last year of this project, we were able to make good progress on the recovery of neurogenic potential of adult neural stem cells after HZE particles irradiation using nestin-CreERT2/R26R-YFP mice. The results are summarized here: 1. Effect of Fe (300 MeV) particles on neurogenic potential of hippocampal adult neural stem cells in irradiated Nestin-CreERT2/R26R-YFP mice: a. One week after Fe particles exposure, tamoxifen (Tam) was given to induce YFP-labeling of adult neural stem cells. Adult hippocampal neurogenesis was analyzed one month after Tam. b. Hippocampal neurogenesis as measured by cells proliferation (BrdU+) and immature neurons (DCX+) was significantly attenuated after 0.5 Gy but not 0.1 Gy of Fe particles in the SGZ of the dentate gyrus. c. Total YFP+ cells or Type-1 neural stem cells in the SGZ were not affected by 0.5 Gy of Fe ions. 2. Recovery of hippocampal neurogenesis were also examined after Oxygen (1000 MeV) particles or X-rays irradiation: a. Oxygen particles up to 1Gy did not cause significant reduction in hippocampal neurogenesis. b. Oxygen particles might enhance hippocampal cell proliferation at lower doses. c. Oxygen particles cause distinctive effect on cell proliferation at the suprablade and the infrablade of dentate gyrus. d. X-rays cause similar effect onto hippocampal neurogenesis as compared to Oxygen particles irradiation. 3. In the past year, we have processed 161 mice at NSRL (11C run) with 96 mice being irradiated by silicon particles (1000 MeV) ranging from 0.2 Gy to 1Gy. After irradiation, all animals were transferred back to UTSW campus. Animal tissue samples were harvested at different time points and were subjected to immunohistochemistry analysis.

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